# Day 32 – Queue Basics

🔷 What is Queue?

Queue ek linear data structure hai jo follow karta hai:

👉 FIFO (First In First Out)
Matlab: jo pehle aayega, wo pehle niklega

📌 Real-life example:

Ticket line

Bus queue

Printer queue

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🔷 Basic Operations of Queue

1. Enqueue (Insert)

Element ko rear (end) me add karna

2. Dequeue (Delete)

Element ko front (start) se remove karna

3. Peek / Front

Front element ko dekhna (remove nahi karna)

4. isEmpty

Queue khaali hai ya nahi

5. isFull

Queue full hai ya nahi (array case me)

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🔷 Queue Representation

📌 Using Array

[10, 20, 30, 40]
↑ ↑
Front Rear

## Java Implementation (Using Array)

```java
package Day_32;

class QueueArray {
    int front, rear, size;
    int arr[];

    QueueArray(int size) {
        this.size = size;
        arr = new int[size];
        front = 0;
        rear = -1;
    }

    // Enqueue
    void enqueue(int x) {
        if (rear == size - 1) {
            System.out.println("Queue Overflow");
            return;
        }
        arr[++rear] = x;
        System.out.println(x + " inserted");
    }

    // Dequeue
    int dequeue() {
        if (front > rear) {
            System.out.println("Queue Underflow");
            return -1;
        }
        int val = arr[front++];
        System.out.println(val + " removed");
        return val;
    }

    // Peek
    int peek() {
        if (front > rear) {
            System.out.println("Queue is Empty");
            return -1;
        }
        return arr[front];
    }

    // Display Queue
    void display() {
        if (front > rear) {
            System.out.println("Queue is Empty");
            return;
        }
        System.out.print("Queue: ");
        for (int i = front; i <= rear; i++) {
            System.out.print(arr[i] + " ");
        }
        System.out.println();
    }
    public static void main(String[] args) {
        QueueArray q = new QueueArray(5);

        q.enqueue(10);
        q.enqueue(20);
        q.enqueue(30);

        q.display();

        q.dequeue();
        q.display();

```

```
        System.out.println("Front element: " + q.peek());
    }
}

```

🔷 Dry Run

Step-by-step:

enqueue(10) → [10]
enqueue(20) → [10,20]
enqueue(30) → [10,20,30]

dequeue() → remove 10
Queue → [20,30]

peek() → 20

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🔷 Time Complexity

| Operation | Time |
|---|---|
| **Enqueue** | O(1) |
| **Dequeue** | O(1) |
| **Peek** | O(1) |

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🔷 Types of Queue

- Simple Queue
- Circular Queue
- Priority Queue
- Deque (Double Ended Queue)

🔷 **Important Interview Points** ⚡

- Queue works on **FIFO**
- Insert → rear, Delete → front
- Array queue me **waste space problem** hota hai
  👉 isliye Circular Queue use hota ha